Kali Fang

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Mechanical Engineering graduate specializing in hardware–software system integration, control systems, and electromechanical design. Hands-on experience developing real-time firmware in C/C++, implementing PID control, and sensor fusion across autonomous robotic platforms. Seeking automation or controls engineering positions.

EDUCATION

Worcester Polytechnic Institute | *Worcester, MA* — Aug 2022 - May 2026
Bachelor of Science in Mechanical Engineering, Minor in Robotics Engineering

EXPERIENCE

Saint-Gobain: CertainTeed Roofing R&D, Glass Mat Co-op - *Northborough, MA* — Jul 2025 - Dec 2025
- Designed and developed Python-based application to automate test results and data acquisition system to visualize data and perform statistical analysis (ANOVA, t-tests, regression), reducing analysis time by 90%
- Developed test protocols and conducted materials characterization using multiple instrumentation systems including SurPASS zeta potential analyzer, NIR spectroscopy, tensile/tear testers, and air permeability systems
- Collaborated with cross-functional teams (chemists, process engineers, manufacturing) to provide data analysis support, conducting statistical evaluations to guide product development decisions
- Authored technical documentation (SOPs, NKC reports) to report and standardize future testing and streamline data collection workflows and ensure repeatability and reproducibility across all users

PROJECTS

Autonomous SLAM Navigation System - *Worcester, MA* — Mar 2026 - Present
- Designed modular ROS2 architecture with custom nodes for frontier detection, path planning, and motion control
- Deployed GMapping-based SLAM pipeline integrating LiDAR data streams to construct and maintain real-time 2D occupancy grids in dynamic environments
- Developed custom A* path planning algorithm with hybrid heuristics incorporating Euclidean distance, distance-to-wall penalties, and angle optimization for safe, efficient navigation
- Utilized OpenMV computer vision to filter and detect obstacles, augmenting sensor fusion inputs for real-time SLAM-based navigation

Accurate and Wearable Blood Pressure Monitor - *Worcester, MA* — Aug 2025 - Present
- Conducted modular, scalable software in C++ using PlatformIO, structuring code for IIR low-pass filtering, oscillometric beat detection, and signal quality assessment for real-time blood pressure results of ±5 mmHg
- Designed and soldered a custom PCB board, integrating analog PPG and pressure sensors with ESP32 microcontroller and H-bridge motor driver
- Designed a custom CAD enclosure to house system components, improving mechanical stability and assembly

Robotic Arm Control System (4-DOF) - *Worcester, MA* — Aug 2024 - Oct 2024
- Worked with a team of 3 engineers to develop real-time motion control system in MATLAB/C++ for 4-DOF robotic manipulator, achieving 90% task accuracy through closed-loop control and camera-based vision feedback
- Developed custom software libraries and optimized control algorithms for kinematics calculations, trajectory planning, and motor control

Autonomous Robotic Maze Navigator - *Worcester, MA* — Mar 2024 - May 2024
- Designed and programmed real-time control system using an ESP32, integrating WiFi communication (MQTT), IMU sensors, and vision recognition system for multi-robot coordination
- Developed multi-threaded architecture for concurrent sensor data processing, path planning, and motor control
- Implemented PID control algorithms for ultrasonic distance sensors and motor control, improving navigation accuracy by 34% through sensor fusion and real-time feedback

SKILLS

Programming Languages: C/C++, C#, Python, MATLAB, JavaScript, SQL
Embedded Systems: Arduino (C/C++), ESP32, Raspberry Pi, PlatformIO, Linux
Control Systems: PID Control, Motion Control, Sensor Fusion, Multi-threading, Real-Time Data Processing
Software: SolidWorks, CAD, FEA, Git/GitHub, Visual Studio, ROS2, Streamlit
Hardware & Testing: Oscilloscope, Logic Analyzer, Stepper/Servo Motor Control, Encoders, Serial Communication (UART, I2C, SPI), Soldering, System Integration